# 178026: muscovite–biotite syenogranite, Russell Well

## Location and sampling

EDMUND (SF 50-14), MANGAROON (2050)
MGA Zone 50, 364600E 7360610N

Sampled on 3 October 2002

The sample was taken from the northeastern edge of a 20 m-long, easterly trending low pavement, 1 km east of Russell Well.

## Tectonic unit/relations

The sample is a pale-grey, coarse-grained, heterogeneous, pegmatitic muscovite–biotite syenogranite that belongs to the 1680–1620 Ma Durlacher Supersuite (Sheppard et al., 2004). The syenogranite is the youngest igneous phase identified at the sampling site, and has intruded a biotite–muscovite granodiorite that is rich in inclusions of augen gneiss and was dated at 1677 ± 5 Ma by Nelson (this compilation; GSWA 178027). At the sampling site, the syenogranite contains irregular layers that are richer in mafic minerals, mainly biotite. It was collected to constrain the age of the Durlacher Supersuite. The data presented below indicate a maximum age of igneous crystallization of 1691 ± 9 Ma for the syenogranite, but it must be younger than 1677 ± 5 Ma, the igneous crystallization age of the granodiorite it intrudes.

## Petrographic description

The sample consists principally of microcline (50 vol.%), quartz (25 vol.%), plagioclase (20 vol.%), muscovite (3 vol.%), and biotite (2 vol.%), with trace amounts of accessory tourmaline, epidote, chlorite, and zircon. This is a heterogeneous muscovite–biotite syenogranite with micaceous and mica-poor domains (?pegmatoidal), some of which contain segregations of tourmaline. The micaceous domains are partly schistose and could be schlieren of metasedimentary origin (probably upper greenschist to lower amphibolite facies), with secondary epidote and rare clear, euhedral zircons. Plagioclase is weakly to moderately saussuritized, with minor epidote and rare chlorite replacing biotite. This is a heterogeneous but mostly K-feldspar-rich granitoid with large domains with and without disseminated biotite. Biotite occurs mostly in a quartz-poor band 10 to 15 mm wide. The bulk mineralogy indicates a two-mica syenogranite with S-type affinities. The plagioclase is present as anhedral and subhedral grains up to 5 mm in length, and locally with a partial mantle of albite exsolved from adjacent grains of microcline. The microcline is mostly anhedral and partly poikilitic, with grains from 2 to 8 mm in diameter. Albite has been exsolved, mostly along grain-boundaries, where "swapped rims" are common. Quartz typically forms weakly deformed larger grains, up to 4 mm long, but areas containing fine-grained recrystallized grains are present, with grains mostly less than 0.4 mm in diameter. Biotite is present as primary flakes, up to 2 mm in length, and aggregates of largely decussate smaller flakes, especially in the mica-rich band that contains most of the muscovite in the thin section. Muscovite is mostly schistose and oriented parallel to the micaceous band, with flakes up to 3 mm long, transecting and post-dating biotite in the same layer. There is also less strongly oriented muscovite in this layer, particularly where it is intergrown with earlier decussate biotite, and the schistosity is not entirely planar, with the possibility of an S–C fabric in places. A 15 mm-diameter patch of tourmaline in the centre of one of the biotite-free areas suggests a pegmatoidal aspect to the those areas. Plagioclase is albitized and shows weak or moderate development of sericite and epidote in a saussuritic alteration pattern. The exsolved plagioclase is mostly fresh, and rarely altered, with fibrous sericite in some areas. Minor epidote has developed in some of the biotite, and rare flakes are altered to chlorite. In the muscovite and biotite lenses there is some replacement of biotite by granular epidote, with lamellae of leucoxene or microcrystalline titanite parallel to the cleavage of the former biotite. There are muscovite and epidote aggregates in the biotite-free areas, but they are not as abundant as in the biotite- and muscovite-rich band. Clear, euhedral zircon crystals up to 70 µm long occur only in the biotite-rich band, with rare zoned crystals containing epidote and metamict-altered ?allanite adjacent to this band. The overall fabric suggests that the micaceous areas contain schlieren of metasedimentary origin as well as disseminated primary magmatic biotite. It is possible that any deformation of this sample was largely partitioned into the micaceous areas; recrystallization of quartz implies amphibolite facies deformation and metamorphism (or autometamorphism). The biotite–muscovite–epidote assemblages in the possible schlieren suggest lower amphibolite to greenschist facies conditions. There is also evidence low-temperature hydrothermal alteration to saussuritic assemblages.

## Zircon morphology

The zircons isolated from this sample are pale yellowish-brown or pinkish-brown, generally between 60 × 100 µm and 100 × 280 µm in size, and euhedral, with some rounded terminations, and slightly elongate. Most grains are structureless, but a minority has faint internal zonation and a few have structureless and fractured cores surrounded by zoned rims. Cathodoluminescence images of representative zircons are given in Figure 1.

## Analytical details

This sample was analysed on 14 July 2003. The counter deadtime during the analysis session was 24 ns. Eight analyses of the CZ3 standard obtained during the analysis session indicated a Pb*/U calibration uncertainty of 1.40% (1σ). Common-Pb corrections were applied

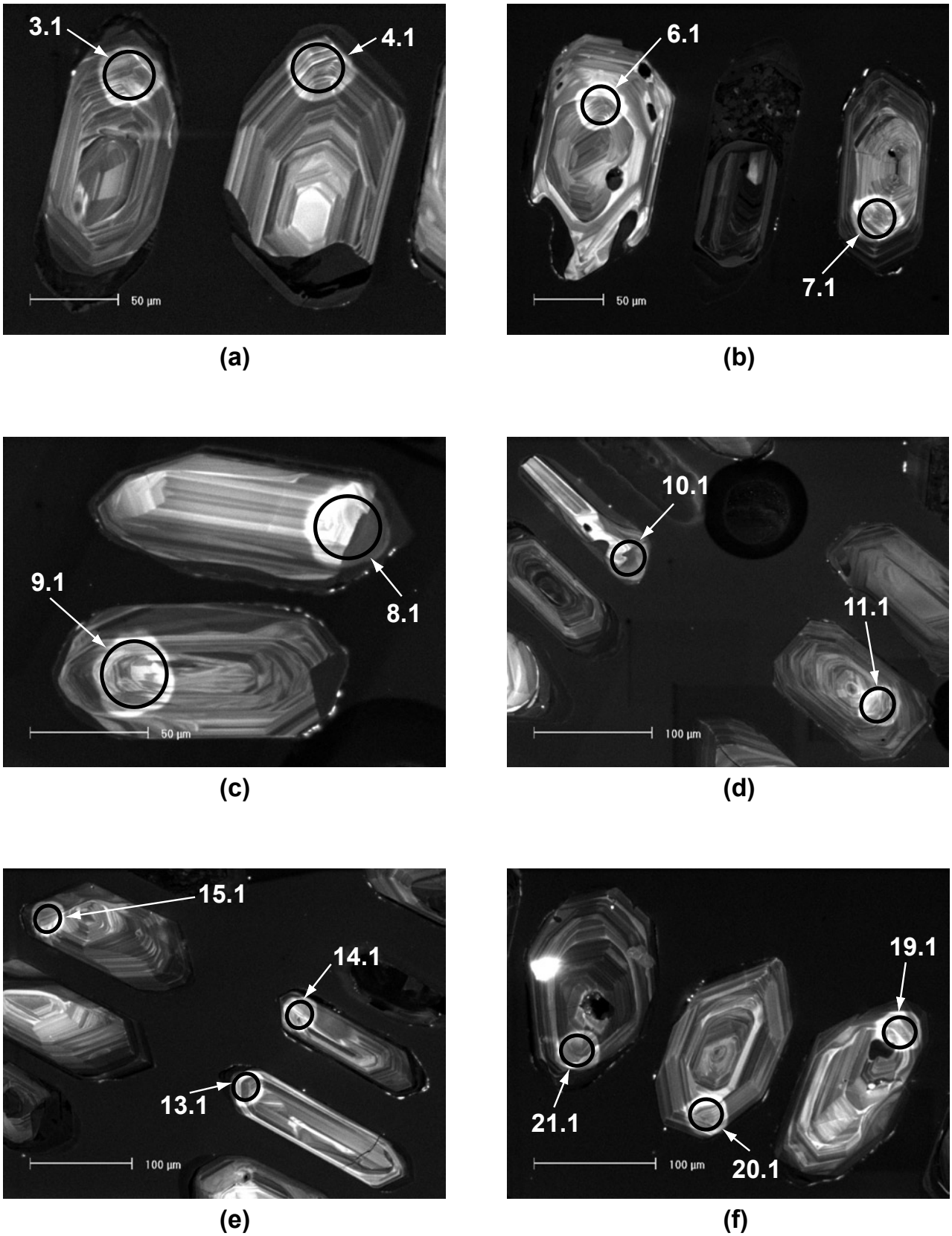


**Figure 1.** **Cathodoluminescence images of representative zircons from sample 178026: muscovite–biotite syenogranite, Russell Well**

**Table 1. Ion microprobe analytical results for zircons from sample 178026: muscovite–biotite syenogranite, Russell Well**

| *Grain .spot* | *U (ppm)* | *Th (ppm)* | *Pb (ppm)* | *f206%* | $^{207}Pb/^{206}Pb$ | $\pm 1\sigma$ | $^{208}Pb/^{206}Pb$ | $\pm 1\sigma$ | $^{206}Pb/^{238}U$ | $\pm 1\sigma$ | $^{207}Pb/^{235}U$ | $\pm 1\sigma$ | *% concordance* | $^{207}Pb/^{206}Pb$ *Age* | $\pm 1\sigma$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.1 | 220 | 156 | 78 | 0.127 | 0.11044 | 0.00079 | 0.20850 | 0.00163 | 0.3107 | 0.0044 | 4.731 | 0.079 | 97 | 1 807 | 13 |
| 2.1 | 495 | 125 | 162 | 0.053 | 0.10967 | 0.00046 | 0.07336 | 0.00063 | 0.3215 | 0.0045 | 4.861 | 0.074 | 100 | 1 794 | 8 |
| 3.1 | 401 | 123 | 133 | 0.054 | 0.10946 | 0.00052 | 0.08930 | 0.00078 | 0.3207 | 0.0045 | 4.841 | 0.075 | 100 | 1 790 | 9 |
| 4.1 | 445 | 122 | 147 | 0.135 | 0.11019 | 0.00052 | 0.07871 | 0.00080 | 0.3212 | 0.0045 | 4.879 | 0.075 | 100 | 1 802 | 9 |
| 5.1 | 313 | 123 | 104 | 0.524 | 0.10948 | 0.00083 | 0.11480 | 0.00163 | 0.3086 | 0.0044 | 4.659 | 0.079 | 97 | 1 791 | 14 |
| 6.1 | 165 | 106 | 60 | 0.294 | 0.10917 | 0.00105 | 0.18405 | 0.00220 | 0.3208 | 0.0046 | 4.829 | 0.088 | 100 | 1 786 | 18 |
| 7.1 | 178 | 93 | 61 | 0.183 | 0.10896 | 0.00093 | 0.15172 | 0.00181 | 0.3134 | 0.0045 | 4.709 | 0.083 | 99 | 1 782 | 16 |
| 8.1 | 402 | 82 | 123 | 0.107 | 0.10501 | 0.00053 | 0.06012 | 0.00077 | 0.3037 | 0.0043 | 4.397 | 0.069 | 100 | 1 714 | 9 |
| 9.1 | 348 | 140 | 117 | 0.061 | 0.11175 | 0.00058 | 0.12124 | 0.00095 | 0.3153 | 0.0045 | 4.858 | 0.076 | 97 | 1 828 | 9 |
| 10.1 | 477 | 35 | 138 | 0.082 | 0.10369 | 0.00051 | 0.02076 | 0.00064 | 0.2980 | 0.0042 | 4.261 | 0.066 | 99 | 1 691 | 9 |
| 11.1 | 258 | 78 | 86 | 0.072 | 0.11057 | 0.00073 | 0.08748 | 0.00120 | 0.3208 | 0.0046 | 4.891 | 0.081 | 99 | 1 809 | 12 |
| 12.1 | 695 | 170 | 234 | 2.080 | 0.10808 | 0.00091 | 0.07055 | 0.00195 | 0.3082 | 0.0044 | 4.594 | 0.080 | 98 | 1 767 | 15 |
| 12.2 | 132 | 176 | 56 | 0.247 | 0.11005 | 0.00108 | 0.38493 | 0.00284 | 0.3265 | 0.0047 | 4.955 | 0.092 | 101 | 1 800 | 18 |
| 13.1 | 296 | 150 | 102 | 0.232 | 0.10942 | 0.00069 | 0.14478 | 0.00130 | 0.3164 | 0.0045 | 4.774 | 0.078 | 99 | 1 790 | 11 |
| 14.1 | 205 | 79 | 69 | 0.298 | 0.10929 | 0.00088 | 0.11215 | 0.00162 | 0.3151 | 0.0045 | 4.748 | 0.082 | 99 | 1 788 | 15 |
| 15.1 | 238 | 51 | 77 | 0.171 | 0.10997 | 0.00076 | 0.06313 | 0.00121 | 0.3172 | 0.0045 | 4.809 | 0.080 | 99 | 1 799 | 13 |
| 16.1 | 116 | 79 | 42 | 0.025 | 0.11049 | 0.00098 | 0.20149 | 0.00197 | 0.3193 | 0.0046 | 4.865 | 0.087 | 99 | 1 808 | 16 |
| 17.1 | 206 | 92 | 70 | 0.138 | 0.10935 | 0.00083 | 0.13127 | 0.00155 | 0.3163 | 0.0045 | 4.769 | 0.081 | 99 | 1 789 | 14 |
| 18.1 | 127 | 60 | 44 | 0.359 | 0.10864 | 0.00121 | 0.13114 | 0.00238 | 0.3185 | 0.0046 | 4.771 | 0.092 | 100 | 1 777 | 20 |
| 19.1 | 162 | 50 | 53 | 0.246 | 0.10857 | 0.00100 | 0.08892 | 0.00183 | 0.3164 | 0.0045 | 4.736 | 0.085 | 100 | 1 776 | 17 |
| 20.1 | 197 | 83 | 66 | 0.130 | 0.11080 | 0.00081 | 0.11972 | 0.00141 | 0.3153 | 0.0045 | 4.817 | 0.081 | 97 | 1 813 | 13 |
| 21.1 | 293 | 206 | 105 | 0.079 | 0.11058 | 0.00063 | 0.20517 | 0.00127 | 0.3148 | 0.0045 | 4.800 | 0.077 | 98 | 1 809 | 10 |
| 22.1 | 118 | 96 | 44 | 0.365 | 0.10960 | 0.00125 | 0.23293 | 0.00275 | 0.3208 | 0.0047 | 4.847 | 0.095 | 100 | 1 793 | 21 |
| 23.1 | 153 | 93 | 55 | 0.237 | 0.10971 | 0.00098 | 0.17876 | 0.00199 | 0.3213 | 0.0046 | 4.861 | 0.087 | 100 | 1 795 | 16 |
| 24.1 | 157 | 106 | 57 | 0.138 | 0.11166 | 0.00093 | 0.20001 | 0.00189 | 0.3183 | 0.0046 | 4.901 | 0.086 | 98 | 1 827 | 15 |

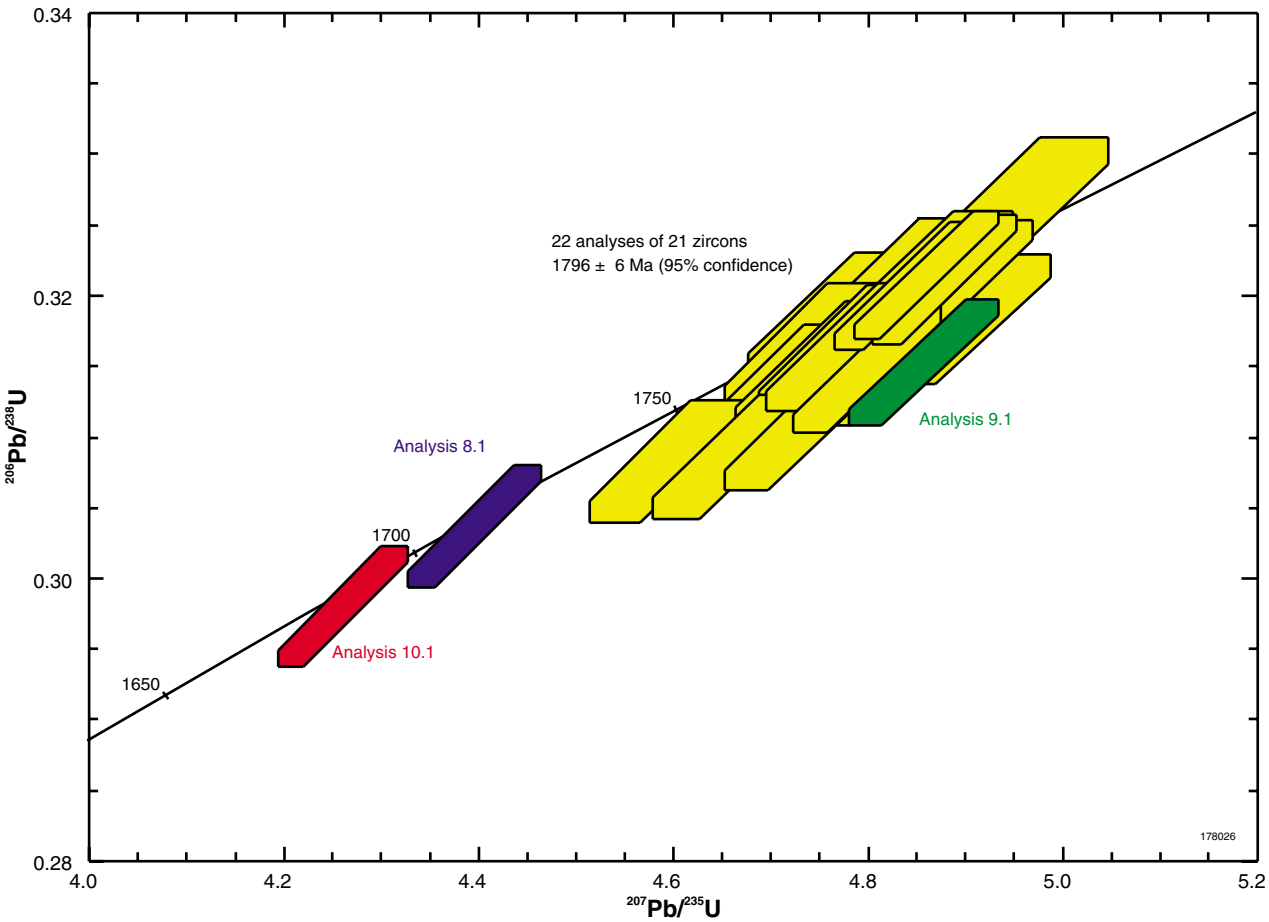


**Figure 2. Concordia plot for sample 178026: muscovite–biotite syenogranite, Russell Well**

assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analysis 12.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

## Results

Twenty-five analyses were obtained from 24 zircons. Results are given in Table 1 and shown on a concordia plot in Figure 2.

## Interpretation

The analyses are concordant to slightly discordant, with the discordance pattern consistent with a recent episode of radiogenic-Pb loss. Twenty-two concordant analyses of 21 zircons have $^{207}$Pb/$^{206}$Pb ratios defining a single population and indicating a weighted mean $^{207}$Pb/$^{206}$Pb date of 1796 ± 6 Ma (chi-squared = 0.86). Concordant analyses 8.1 and 10.1 indicate significantly younger $^{207}$Pb/$^{206}$Pb dates, whereas discordant analysis 9.1 indicates a slightly older $^{207}$Pb/$^{206}$Pb date, than the main population.

The distribution of analyses along the concordia curve indicates that most, if not all, of the zircons within this sample are xenocrysts. Analysis 10.1 was obtained on a 30 × 100 µm, clear, structureless grain. The date of 1691 ± 9 Ma (±1σ uncertainty) indicated by the $^{207}$Pb/$^{206}$Pb ratio of concordant analysis 10.1 is therefore interpreted as the maximum age for igneous crystallization of the syenogranite.

Recommended reference for this publication:

NELSON, D. R., 2005, 178026: muscovite-biotite syenogranite, Russell Well; Geochronology dataset 535; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 14/07/2003; Data released: 30/06/2005